

In-Depth Technical Guide to Etacelasil (CAS No. 37894-46-5)

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Compound of Interest

Compound Name: Etacelasil
CAS No.: 37894-46-5
Cat. No.: B1216580

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Abstract

Etacelasil (CAS No. 37894-46-5), an organosilicon compound, is recognized for its role as a plant growth regulator. Its primary mechanism of action involves the controlled release of ethylene, a key phytohormone that governs a wide array of physiological processes in plants. This technical guide provides a comprehensive overview of **Etacelasil**, including its chemical properties, mechanism of action, and its application in agriculture. A detailed exploration of the ethylene signaling pathway, which is activated by the ethylene released from **Etacelasil**, is also presented. While specific quantitative data on **Etacelasil**'s hydrolysis kinetics and ethylene release rates are not extensively available in publicly accessible literature, this guide synthesizes the existing knowledge to provide a thorough understanding of its function and application.

Chemical and Physical Properties

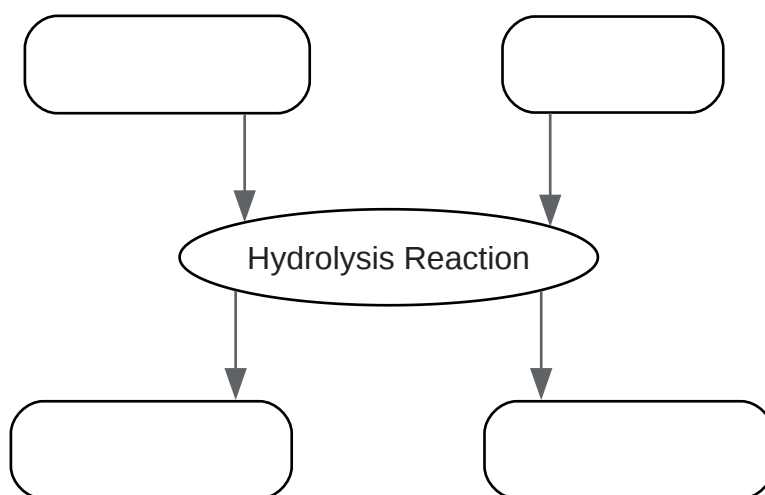
Etacelasil is chemically known as 6-(2-chloroethyl)-6-(2-methoxyethoxy)-2,5,7,10-tetraoxa-6-silaundecane. Its fundamental properties are summarized in the table below.

Property	Value
CAS Number	37894-46-5
Molecular Formula	C ₁₁ H ₂₅ ClO ₆ Si
Molecular Weight	316.85 g/mol
IUPAC Name	2-chloroethyl-tris(2-methoxyethoxy)silane
Synonyms	Alsol, CGA 13586
Appearance	Liquid (at room temperature)

Mechanism of Action: Ethylene Release via Hydrolysis

The primary function of **Etacelasil** as a plant growth regulator stems from its ability to release ethylene gas upon hydrolysis. The presence of water initiates a chemical breakdown of the **Etacelasil** molecule, yielding ethylene and other byproducts.

This controlled release of ethylene allows for the targeted manipulation of various plant developmental processes. The rate of ethylene release is influenced by environmental factors such as moisture and pH.



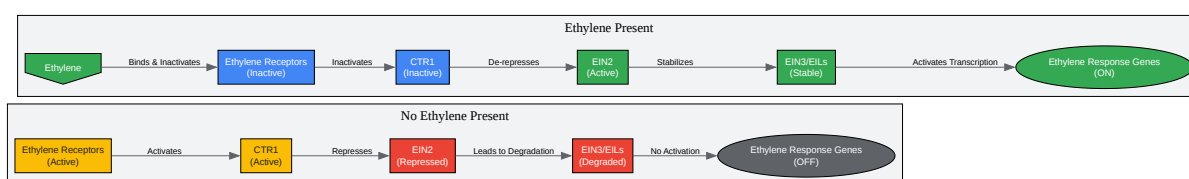
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Caption: Hydrolysis of **Etacelasil** to release ethylene.

The Ethylene Signaling Pathway in Plants

The ethylene released from **Etacelasil** initiates a well-defined signaling cascade within plant cells. This pathway ultimately leads to changes in gene expression and the physiological responses associated with ethylene. In the absence of ethylene, its receptors, located on the endoplasmic reticulum, activate a protein kinase called CTR1 (CONSTITUTIVE TRIPLE RESPONSE 1). CTR1, in turn, represses a downstream protein called EIN2 (ETHYLENE INSENSITIVE 2).

When ethylene is present, it binds to the receptors, inactivating them. This inactivation of the receptors leads to the deactivation of CTR1. With CTR1 inactive, the repression on EIN2 is lifted. EIN2 is then cleaved, and a fragment of it translocates to the nucleus. In the nucleus, this EIN2 fragment stabilizes transcription factors such as EIN3 (ETHYLENE INSENSITIVE 3) and EILs (EIN3-like proteins). These stabilized transcription factors then regulate the expression of a host of ethylene-responsive genes, leading to the observed physiological effects.



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Caption: The canonical ethylene signaling pathway in plants.

Applications in Agriculture and Research

Etacelasil's ability to induce ethylene-related responses has led to its use in various agricultural applications to manipulate plant growth and development.

Fruit Ripening

Ethylene is a key hormone in the ripening of climacteric fruits such as bananas, tomatoes, and avocados. While specific protocols for **Etacelasil** are not widely published, the principle involves applying an **Etacelasil** solution to mature, unripe fruits to initiate and synchronize the ripening process.

Fruit Abscission

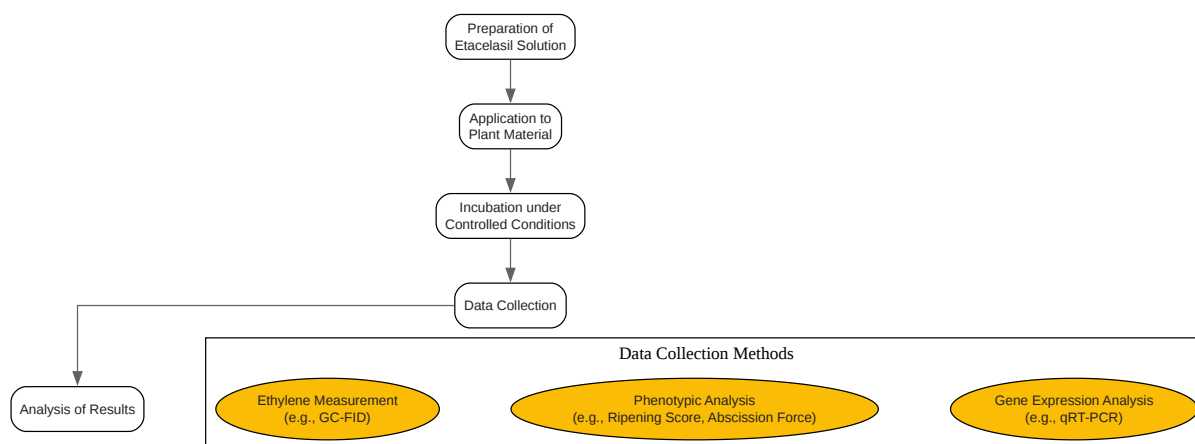
In mechanical harvesting, reducing the force required to remove fruits from the plant is crucial. Ethylene promotes the formation of an abscission layer at the base of the fruit stem. Research has shown that **Etacelasil**, under the name Alsol, is effective in reducing the fruit removal force in olives, potentially more so than other ethylene-releasing compounds like ethephon.^[1]

Induction of Flowering

In some plant species, such as pineapples, ethylene can induce flowering. While direct experimental protocols for **Etacelasil** in this application are scarce, the analogous use of acetylene gas (generated from calcium carbide) to force pineapple flowering illustrates the potential for ethylene-releasing compounds to control flowering time.

Experimental Protocols and Methodologies

Detailed, standardized experimental protocols for the application of **Etacelasil** are not readily available in the public domain. However, based on the use of other ethylene-releasing agents and general practices in plant science, the following outlines a general workflow for studying the effects of **Etacelasil**.



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Caption: General experimental workflow for **Etacelasil** studies.

Protocol Considerations:

- **Concentration:** The optimal concentration of **Etacelasil** will vary depending on the plant species, the developmental stage, and the desired outcome. Pilot studies are essential to determine the effective concentration range.
- **Application Method:** Application can be via spraying, dipping, or injection, depending on the plant material and experimental setup.
- **Control Groups:** Appropriate control groups (e.g., untreated, water-treated) are crucial for interpreting the results.

- Environmental Conditions: Temperature and humidity can affect the rate of hydrolysis and ethylene release, and should be carefully controlled and monitored.

Quantitative Analysis of Etacelasil and Its Residues

The analysis of **Etacelasil** residues in plant tissues is important for regulatory purposes and for understanding its uptake and metabolism. Techniques such as liquid chromatography-tandem mass spectrometry (LC-MS/MS) are well-suited for the sensitive and selective quantification of pesticide residues in complex matrices like plant material.

General Analytical Method:

- Extraction: Homogenized plant material is extracted with an organic solvent (e.g., acetonitrile).
- Cleanup: The extract is purified using techniques like QuEChERS (Quick, Easy, Cheap, Effective, Rugged, and Safe) to remove interfering matrix components.
- Analysis: The cleaned extract is analyzed by LC-MS/MS.

Conclusion

Etacelasil is a valuable tool in plant science research and agriculture due to its function as an ethylene-releasing agent. Its ability to initiate ethylene-dependent physiological processes offers a means to control fruit ripening, abscission, and flowering. While a detailed quantitative understanding of its hydrolysis and ethylene release kinetics is an area requiring further public research, the well-established knowledge of the ethylene signaling pathway provides a solid framework for understanding its mode of action. Future research should focus on establishing standardized protocols and generating quantitative data to optimize the application of **Etacelasil** for various agricultural purposes.

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References

- 1. volcaniarchive.agri.gov.il [volcaniarchive.agri.gov.il]
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